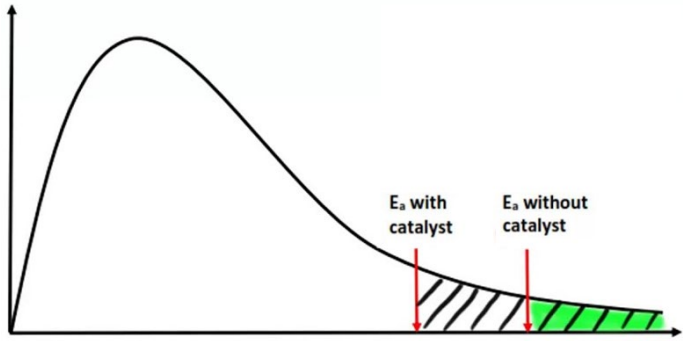


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			at higher concentration, more particles in a given volume / particles closer together (1) more frequent collisions / more chance of collisions (1)	2			2		
	(b)	(i)		using expts 1 and 2 - when [NO] doubled (1) rate increases by factor of 4 / rate $\propto$ [NO] ² (1) or using expts 1 and 3 - when [NO] trebled (1) rate increases by factor of 9 / rate $\propto$ [NO] ² (1)			2	2	2	
		(ii)		using expts 2 and 4 when [O ₂] doubled (1) rate stays the same / rate unaffected by [O ₂] (1)			2	2	2	
		(iii)		monitor changes in volume of gas / use a gas syringe / monitor changes in pressure / use a manometer (1) reagents have more moles of gas (1) or use a colorimeter (1) since NO ₂ is brown/coloured (1)			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		award (1) each for any two of following • NO/NO₂/CO are toxic • CO₂ contributes to greenhouse effect / global warming • NO₂/NO_x contribute to acid rain		2		2		
		(v)		exhaust (1) award (1) for any of following • heterogenous • a transition metal / a transition metal compound • palladium / platinum / rhodium	2			2		
	(c)			 shape of curve (1) lower activation energy with catalyst / shown on graph (1) more collisions / molecules have energy greater than activation energy (1)	3			3		
				Question 8 total	7	2	6	15	4	2